



File Processing



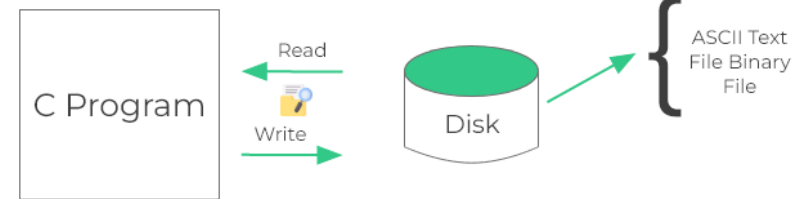
File Introduction

- Files are places where data can be stored permanently.
- Some programs expect the same set of data to be fed as input every time it is run.
 - ❑ Cumbersome.
 - ❑ Better if the data are kept in a file, and the program reads from the file.

Text File vs Binary File

Based on “ASCII code”

File Handling in C



Text Files

- When using a **text** file, we write out separately each of the pieces of data about a given record.
- The text file will be **readable** by an editor

Binary Files

- When using a **binary** file we write whole record data to the file at once.
- but the numbers in the binary file will **not be readable** in this way.

Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040	 	Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	\$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174	|	
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177		DEL

ASCII Code	Char	155	¢	185	ƒ	218	ƒ
128	Ç	156	£	186		219	■
129	ü	157	¥	187	┘	220	■
130	é	158	Pts	188	┘	221	┘
131	â	159	f	189	┘	222	┘
132	ä	160	á	190	┘	223	■
133	à	161	í	191	┘	224	α
134	å	162	ó	192	┘	225	β
135	ç	163	ú	193	┘	226	Γ
136	ê	164	ñ	194	┘	227	π
137	ë	165	Ñ	195	┘	228	Σ
138	è	166	a	196	—	229	σ
139	ï	167	o	197	┘	230	μ
140	î	168	¿	198	┘	231	τ
141	ì	169	¡	199	┘	232	Φ
142	Ä	170	¬	200	┘	233	Θ
143	Å	171	½	201	┘	234	Ω
144	É	172	¼	202	┘	235	δ
145	§	173	ı	203	┘	236	∞
146	Æ	174	«	204	┘	237	φ
147	ô	175	»	205	┘	238	ε
148	ö	176	⋮	206	┘	239	∩
149	ò	177	⋮	207	┘	240	≡
150	û	178	■	208	┘	241	±
151	ù	179	┘	209	┘	242	≡
152	ÿ	180	┘	210	┘	243	≡
153	Ö	181	┘	211	┘	244	┘
154	Ü	182	┘	212	┘	245	┘
		183	π	213	┘	246	÷
		184	ƒ	214	┘	247	≈
				215	┘	248	°
				216	┘		
				217	┘		

249	•
250	•
251	✓
252	n
253	2
254	■
255	



File Edit Options Encoding Help

```

line-1
line-2
line-3
line-4

---- SAMPLE 1
#include <iostream>

int main()
{
char ch;
FILE *fp1 = fopen( "D:/mydoc/cos2101/DemoCOS21
if (fp1 == NULL) {
printf("cannot open file !!");
return 0;
}
do {
long pos = ftell(fp1);
ch = getc(fp1);
printf("%c \t %x \t %d \n",ch, ch , pos);
} while (ch != EOF);

fclose(fp1);

}

```

TEXT FILE

File Edit Options Encoding Help

```

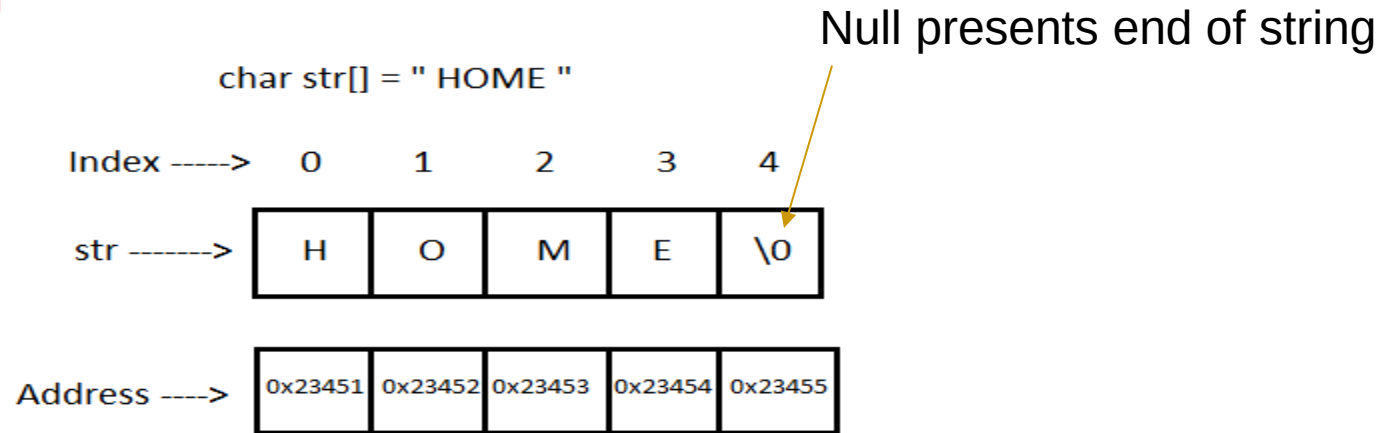
00000000: 6C 69 6E 65 2D 31 0D 0A|6C 69 6E 65 2D 32 0D 0A| line-1..line-2..
00000010: 6C 69 6E 65 2D 33 0D 0A|6C 69 6E 65 2D 34 0D 0A| line-3..line-4..
00000020: 0D 0A 2D 2D 2D 20 53 41|4D 50 4C 45 20 31 0D 0A| ..--- SAMPLE 1..
00000030: 23 69 6E 63 6C 75 64 65|20 3C 69 6F 73 74 72 65| #include <iostre
00000040: 61 6D 3E 0D 0A 0D 0A 69|6E 74 20 6D 61 69 6E 28| am>....int main(
00000050: 29 0D 0A 7B 0D 0A 63 68|61 72 20 63 68 3B 0D 0A| )..{..char ch;..
00000060: 46 49 4C 45 20 2A 66 70|31 20 3D 20 66 6F 70 65| FILE *fp1 = fope
00000070: 6E 28 20 22 44 3A 2F 6D|79 64 6F 63 2F 63 6F 73| n( "D:/mydoc/cos
00000080: 32 31 30 31 2F 44 65 6D|6F 43 4F 53 32 31 30 31| 2101/DemoCOS2101
00000090: 2F 44 65 62 75 67 2F 74|65 73 74 31 2E 74 78 74| /Debug/test1.txt
000000A0: 22 2C 20 22 72 74 22 29|3B 0D 0A 69 66 20 28 66| ", "rt");..if (f
000000B0: 70 31 20 3D 3D 20 4E 55|4C 4C 29 20 7B 0D 0A 70| p1 == NULL) {..p
000000C0: 72 69 6E 74 66 28 22 63|61 6E 6E 6F 74 20 6F 70| rntf("cannot op
000000D0: 65 6E 20 66 69 6C 65 20|21 21 22 29 3B 0D 0A 72| en file !!");..r
000000E0: 65 74 75 72 6E 20 30 3B|0D 0A 7D 0D 0A 64 6F 20| eturn 0;..}..do
000000F0: 7B 0D 0A 6C 6F 6E 67 20|70 6F 73 20 3D 20 66 74| {..long pos = ft
00000100: 65 6C 6C 28 66 70 31 29|3B 0D 0A 63 68 20 3D 20| ell(fp1);..ch =
00000110: 67 65 74 63 28 66 70 31|29 3B 0D 0A 70 72 69 6E| getc(fp1);..prin
00000120: 74 66 28 22 25 63 20 5C|74 20 20 25 78 20 5C 74| tf("%c \t %x \t
00000130: 20 25 64 20 5C 6E 22 2C|63 68 2C 20 63 68 20 2C| %d \n",ch, ch ,
00000140: 20 70 6F 73 29 3B 0D 0A|7D 20 77 68 69 6C 65 20| pos);..} while
00000150: 28 63 68 20 21 3D 20 45|4F 46 29 3B 0D 0A 0D 0A| (ch != EOF);....
00000160: 66 63 6C 6F 73 65 28 66|70 31 29 3B 0D 0A 0D 0A| fclose(fp1);....
00000170: 7D 0D 0A 0D 0A 2D 20 20|53 41 4D 50 4C 45 20 32| }....-- SAMPLE 2
00000180: 0D 0A 23 64 65 66 69 6E|65 20 5F 43 52 54 5F 53| ..#define _CRT_S
00000190: 45 43 55 52 45 5F 4E 4F|5F 57 41 52 4E 49 4E 47| ECURE_NO_WARNING
000001A0: 53 0D 0A 23 69 6E 63 6C|75 64 65 20 3C 69 6F 73| S..#include <ios
000001B0: 74 72 65 61 6D 3E 0D 0A|0D 0A 0D 0A 69 6E 74 20| tream>.....int
000001C0: 6D 61 69 6E 28 29 0D 0A|7B 0D 0A 63 68 61 72 20| main()..{..char
000001D0: 63 68 3B 0D 0A 46 49 4C|45 20 2A 66 70 31 20 3D| ch;..FILE *fp1 =
000001E0: 20 66 6F 70 65 6E 28 22|44 3A 2F 6D 79 64 6F 63| fopen("D:/mydoc
000001F0: 2F 63 6F 73 32 31 30 31|2F 44 65 6D 6F 43 4F 53| /cos2101/DemoCOS
00000200: 32 31 30 31 2F 44 65 62|75 67 2F 74 65 73 74 31| 2101/Debug/test1
00000210: 2E 74 78 74 22 2C 20 22|72 74 22 29 3B 0D 0A 69| .txt", "rt");..i
00000220: 66 20 28 66 70 31 20 3D|3D 20 4E 55 4C 4C 29 20| f (fp1 == NULL)
00000230: 7B 0D 0A 70 72 69 6E 74|66 28 22 63 61 6E 6E 6F| {..printf("canno
00000240: 74 20 6F 70 65 6E 20 66|69 6C 65 20 21 21 22 29| t open file !!");
00000250: 3B 0D 0A 72 65 74 75 72|6E 20 30 3B 0D 0A 7D 0D| ;..return 0;..}
00000260: 0A 0D 0A 64 6F 20 7B 0D|0A 6C 6F 6E 67 20 70 6F| ...do {..long po

```

[illegible]

FILE CONCEPT

■ Array of character



```
int index = 0;
```

```
char c1 = str[ index ];  
index++;
```

```
char c2 = str[ index ];  
index++;
```

```
str[ index ] = 'A'
```

index identifies location of elements on array to do such as read and write, called access.

read

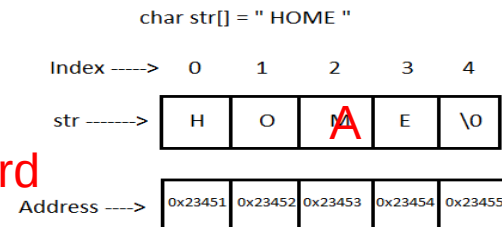
adjust position forward

read

adjust position forward

write

adjust position backward



FILE CONCEPT

```
int index = 0;
char str[100];
char getc(int &index )
{
    char c = str[index];
    index++;
    return c;
}
```

```
    openc();
    char c1 = getc()
    char c2 = getc()
    cout << tell();
    putc( 'A'
    cout << tell();
```

```
void putc(char c )
{
    str[index] = c;
}
```

```
int i openc()
{
    strcpy(str,"HOME");
    return index;
}
```

```
int tell()
{
    return index;
}
```

char str[] = " HOME "

Index ----> 0 1 2 3 4

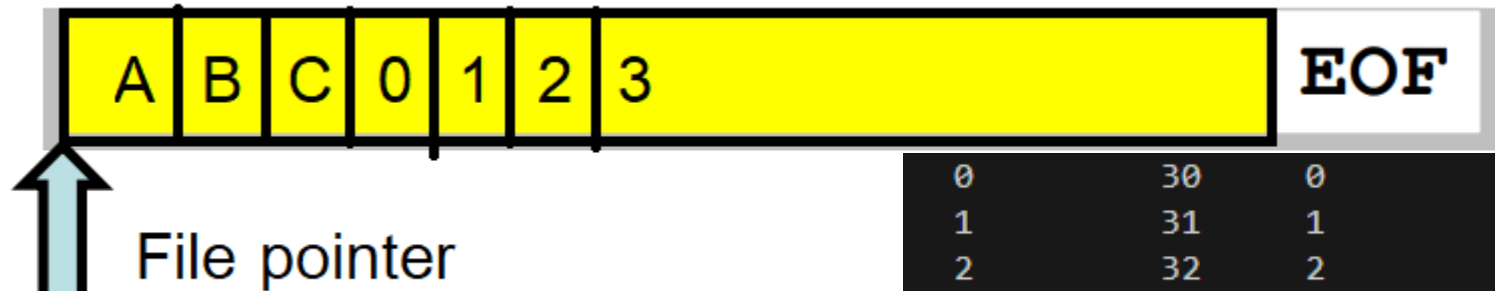
str ----->

H	O	M	E	\0
---	---	---	---	----

Address ---->

0x23451	0x23452	0x23453	0x23454	0x23455
---------	---------	---------	---------	---------

File could be viewed as array of character



```
3 int main()
4 {
5     char ch;
6     FILE *fp1 = fopen( "d:/testLine.txt", "rt");
7     if (fp1 == NULL) {
8         printf("cannot open file !!");
9         return 0;
10    }
11    do {
12        long pos = ftell(fp1);
13        ch = getc(fp1);
14        printf("%c \t %x \t %d \n",ch, ch , pos);
15    } while (ch!=EOF);

16
17    printf("\ndata @pos 3\n");
18    fseek(fp1,3,SEEK_SET);
19    long pos = ftell(fp1);
20    ch = getc(fp1);
21    printf("%c \t %x \t %d \n",ch, ch , pos);
22    fclose(fp1);
23
24 }
```

0	30	0
1	31	1
2	32	2
3	33	3
4	34	4
5	35	5
6	36	6
7	37	7
8	38	8
9	39	9
0	30	10
a	61	11
b	62	12
c	63	13
d	64	14
e	65	15
f	66	16
g	67	17
␣	ffffff	18

data @pos 3

3	33	3
---	----	---

PS D:\mydoc\cos2101\apirak2\lab\1

ลำดับการทำงานแฟ้มข้อมูล

1. เปิดแฟ้มข้อมูล

1. fopen ()

2. การจัดการบนแฟ้มข้อมูล

2.1. อ่าน

2.1. fgetc(), fscanf() , fgets() , fread()

2.2. เขียน

2.2. fputc(), fprintf() , fputs() , fwrite()

2.3. การสืบค้นตำแหน่ง

2.3. fseek() , ftell ()

3. ปิดแฟ้มข้อมูล

3. fclose()

Text File

ลำดับ operation บน text file

1. เปิดแฟ้มข้อมูล

1. fopen ()

2. การจัดการบนแฟ้มข้อมูล

2.1. อ่าน

2.1. fgetc(), fscanf() , fgets()

2.2. เขียน

2.2. fputc(), fprintf() , fputs()

2.3. การสื่บค้นตำแหน่ง

3. ปิดแฟ้มข้อมูล

3. fclose()

Opening a File

- A file must be “opened” before it can be used.

```
FILE *fp;  
:  
fp = fopen (filename, mode);
```

- **fp** is declared as a pointer to the data type FILE.
- **filename** is a string - specifies the name of the file.
- **fopen** returns a pointer to the file which is used in all subsequent file operations.
- **mode** is a string which specifies the purpose of opening the file:
 - “r” :: open the file for reading only
 - “w” :: open the file for writing only
 - “a” :: open the file for appending data to it

MODES

- ***r*** - open a file in read-mode, set the pointer to the beginning of the file.
- ***w*** - open a file in write-mode, set the pointer to the beginning of the file.
- ***a*** - open a file in write-mode, set the pointer to the end of the file.
- ***rb*** - open a binary-file in read-mode, set the pointer to the beginning of the file.
- ***wb*** - open a binary-file in write-mode, set the pointer to the beginning of the file.
- ***ab*** - open a binary-file in write-mode, set the pointer to the end of the file.
- ***r+*** - open a file in read/write-mode, if the file does not exist, it will not be created.
- ***w+*** - open a file in read/write-mode, set the pointer to the beginning of the file.
- ***a+*** - open a file in read/append mode.
- ***r+b*** - open a binary-file in read/write-mode, if the file does not exist, it will not be created.
- ***w+b*** - open a binary-file in read/write-mode, set the pointer to the beginning of the file.
- ***a+b*** - open a binary-file in read/append mode.

Contd.

■ Points to note:

- ❑ Several files may be opened at the same time.
- ❑ For the “w” and “a” modes, if the named file does not exist, it is automatically created.
- ❑ For the “w” mode, if the named file exists, its contents will be overwritten.

Examples

```
FILE  *in, *out ;  
in = fopen ("mydata.dat", "r") ;  
out = fopen ("result.dat", "w");
```

```
FILE  *empl ;  
char filename[25];  
scanf ("%s", filename);  
empl = fopen (filename, "r") ;
```

Closing a File

- After all operations on a file have been completed, it must be closed.
 - Ensures that all file data stored in memory buffers are properly written to the file.
- General format: `fclose (file_pointer) ;`
`FILE *xyz ;`
`xyz = fopen ("test.txt", "w") ;`
`.....`
`fclose (xyz) ;`

Contd

- ❑ **fclose (*FILE* pointer)**
 - Closes specified file
 - Performed automatically when program ends
 - Good practice to close files explicitly
 - system resources are freed.
 - Also, you might not find that all the information that you've written to the file has actually been written to disk until the file is closed.
- ❑ **feof (*FILE* pointer)**
 - Returns true if end-of-file indicator (no more data to process) is set for the specified file

Files and Streams

- Read/Write functions in standard library
 - **getc**
 - Reads one character from a file
 - Takes a **FILE** pointer as an argument
 - `fgetc( stdin )` equivalent to `getchar()`
 - **putc**
 - Writes one character to a file
 - Takes a **FILE** pointer and a character to write as an argument
 - `fputc( 'a', stdout )` equivalent to `putchar( 'a' )`
 - **scanf / fprintf**
 - File processing equivalents of **scanf** and **printf**

Read/Write Operations on Files

- The simplest file input-output (I/O) functions are **getc** and **putc**.
- **getc** is used to read a character from a file and return it.

```
char ch; FILE *fp;
```

```
.....
```

```
ch = getc (fp) ;
```

- **getc** will return an end-of-file marker EOF, when the end of the file has been reached.

- **putc** is used to write a character to a file.

```
char ch; FILE *fp;
```

```
.....
```

```
putc (ch, fp) ;
```

Example :: convert a text file to all UPPERCASE

```
1  #define _CRT_SECURE_NO_WARNINGS
2  #include <iostream>
3
4  int main()
5  {
6      FILE  *in, *out;
7      char  c1;
8
9      in = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/test1.txt", "r");
10     out = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/test1.out", "w");
11
12     while ((c1 = getc(in)) != EOF) {
13         char c2 = toupper(c1);
14         putc(c2, out);
15         printf("%c", c2);
16     }
17     fclose(in);
18     fclose(out);
19
20 }
21
```

Contd.

- We can also use the file versions of **scanf** and **printf**, called **fscanf** and **fprintf**.

- General format:

fscanf (file_pointer, control_string, list) ;

fprintf (file_pointer, control_string, list) ;

- Examples:

fscanf (fp, “%d %s %f”, &roll, dept_code, &cgpa) ;

fprintf (out, “\nThe result is: %d”, xyz) ;

Contd.

`fprintf`

- Used to print to a file
- It is like `printf`, except first argument is a **FILE** pointer (pointer to the file you want to print in)

Some Points

- How to check EOF condition when using **fscanf**?

- Use the function **fEOF**

```
if (fEOF (fp))
```

```
printf ("\n Reached end of file") ;
```

- How to check successful open?

- For opening in “r” mode, the file must exist.

```
if (fp == NULL)
```

```
printf ("\n Unable to open file") ;
```

```

8  int main()
9  {
10     FILE    *in, *out;
11     char     c1;

12
13     in = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/format.txt", "rt");
14     out = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/format.out", "w");
15
16     while ( !feof(in) ) {
17         //line-1 10 c 10.5 text-info-1
18         char txt1[100];
19         int i1;
20         char c1;
21         float d1;
22         char txt2[100];
23         fscanf(in,"%s %d %c %f %s\n", txt1, &i1, &c1, &d1, txt2);
24         fprintf(out, "%s %d %c %f %s\n", txt1, i1, c1, d1, txt2);
25         fprintf(stdout, "%s %d %c %f %s\n", txt1, i1, c1, d1, txt2);
26     }
27     fclose(in);
28     fclose(out);
29
30 }

```

```

86  /*
87  line-1 10 c 10.5 text-info-1
88  line-2 20 c 20.51 text-info-2
89  line-3 30 c 31.5 text-info-3
90  */

```

Operation on File

- Read Only
- Create a new File or over write exist File
- Read/Write File (update file)
- Append File

```

4 int main()
5 {
6     FILE *fileA, /* first input file */
7         *fileB, /* second input file */
8         *fileC; /* output file to be created */
9     int num1, /* number to be read from first file */
10        num2; /* number to be read from second file */
11     int f1, f2;
12
13     /* Open files for processing */
14     fileA = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/class1.txt", "r");
15     fileB = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/class2.txt", "r");
16     fileC = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/class.txt", "w");
17
18     /* As long as there are numbers in both files,
19        read and compare numbers one by one.
20        Write the smaller number to the output file and read the next number
21        */
22     f1 = fscanf(fileA, "%d", &num1);
23     f2 = fscanf(fileB, "%d", &num2);
24     while ((f1 != EOF) && (f2 != EOF)) {
25         if (num1 < num2) {
26             fprintf(fileC, "%d\n", num1);
27             f1 = fscanf(fileA, "%d", &num1);
28         }
29         else if (num2 < num1) {
30             fprintf(fileC, "%d\n", num2);
31             f2 = fscanf(fileB, "%d", &num2);
32         }
33         else { /* numbs are equal: read from both files */
34             fprintf(fileC, "%d\n", num1);
35             f1 = fscanf(fileA, "%d", &num1);
36             f2 = fscanf(fileB, "%d", &num2);
37         }
38     }
39 }

```

Lister - [

File Edit (

1
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12

File Edi

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12
14
16

```
1  #include <iostream>
2  #include <stdio.h>
3  int main()
4  {
5      FILE * pFileTXT;
6      int counter;
7      char cstr[]="abcdefghijk";
8      char istr[]="0123456789";
9      // use "a" for append,
10     //"w" to overwrite, previous content will be deleted
11     pFileTXT = fopen ("d:/aTextFile.txt","a");
12     for(counter=0;counter< 8 ;counter++) {
13         fprintf (pFileTXT, "%c", cstr[counter] );
14     }
15     fprintf(pFileTXT,"\n");// newline
16
17     for(counter=0;counter< 9; counter++) {
18         fprintf (pFileTXT, "%d", istr[counter] );
19     }
20     fprintf(pFileTXT,"A Sentence");
21     fprintf(pFileTXT,"\n"); // newline
22
23     fclose (pFileTXT); // must close after opening
24     return 0;
25 }
```

จากปัญหา

ตัวอย่าง 1

สมมติว่าข้อมูล อาจารย์คณะวิทยาศาสตร์มี 150 ดังนี้

เพศ (sex) : ชาย = 1 , หญิง = 2

อายุ (age) : ? ปี

สถานภาพสมรส (status) : โสด = 1 , แต่งงาน = 2 , อื่นๆ = 2

ปริญญาสูง (degree) : ตรี = 1 , โท = 2 , เอก = 3

จำนวนปีที่สอน (experience) : ? ปี

จงเขียน

1. โปรแกรมรับข้อมูลของอาจารย์คณะวิทยาศาสตร์จำนวน 150 คน

2. หาจำนวนปีเฉลี่ยของ อายุ และ จำนวนปีที่สอน

AVEREAGE AGE = xx.xx Years

AVEREAGE Experience = XX.XX Years

```
1  #include <iostream>
2  using namespace std;
3  struct employee {
4      int sex;
5      int age;
6      int stat;
7      int degr;
8      int tyear;
9  };
10 int main( )
11 {
12     struct employee person[150];
13     int i ;
14     float avg_age,avg_expr;
15     for ( i = 0 ; i < 150 ; i++ ) {
16         cout << "sex = ";
17         cin >> person[i].sex;
18         cout << "age = ";
19         cin >> person[i].age ;
20         cout << "stat = ";
21         cin >> person[i].stat;
22         cout << "degr = ";
23         cin >> person[i].degr;
24         cout << "tyear = ";
25         cin >> person[i].tyear;
26     }
27     avg_age = 0 ;    avg_expr = 0;
28     for ( i = 0 ; i < 150 ; i++ )
29     {
30         avg_age = avg_age + person[i].age ;
31         avg_expr = avg_expr + person[i].tyear ;
32     }
33     avg_age = avg_age/150 ;
34     avg_expr = avg_expr /150 ;
35
36     cout << "AVG AGE = " << avg_age << "\n" ;
37     cout << "AVG EXPR = " << avg_expr << "\n";
38     return 0;
39 }
```

ตัวอย่าง 1

สมมติว่าข้อมูล อาจารย์คณะวิทยาศาสตร์มี 150 ดังนี้

เพศ (sex) : ชาย = 1 , หญิง = 2

อายุ (age) : ? ปี

สถานภาพสมรส (status) : โสด = 1 , แต่งงาน = 2 , อื่นๆ = 2

ปริญญาสูง (degree) : ตรี = 1 , โท = 2 , เอก = 3

จำนวนปีที่สอน (experience) : ? ปี

จงเขียน

1. โปรแกรมรับข้อมูลของอาจารย์คณะวิทยาศาสตร์จำนวน 150 คน

2. หาจำนวนปีเฉลี่ยของ อายุ และ จำนวนปีที่สอน

AVERAGE AGE = XX.XX Years

AVERAGE Experience = XX.XX Years

T65001001, รหัสอาจารย์ (2)

1, เพศ

20, จำนวนปีที่สอน

1, ระดับปริญญา

1 สถานภาพสมรส

ให้นักศึกษา เขียนโปรแกรมอ่านข้อมูลอาจารย์จากแฟ้ม (file) ซึ่งมีรูปแบบ (format) การจัดเก็บดังนี้

T65001001,1,20,1,1

T65001002,2,22,2,2

T65001003,1,21,2,2

T65001004,2,22,1,2

T65001005,2,23,3,2

T65001006,2,19,3,2

```

23 struct employee {
24     char Tcode[10];
25     int sex;
26     int age;
27     int stat;
28     int degr;
29     int tyear;
30 };
31 int main()
32 {
33     struct employee person[150];
34     int i=0;
35     float avg_age, avg_expr;
36     //get data
37
38     FILE *fileA;
39
40     /* Open files for processing */
41     fileA = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/teacherInfo.txt", "rt");
42
43     cout << "code" << setw(10) << "sex" << setw(4)
44         << "year" << setw(4) << "degr" << setw(4) << "status" << endl;
45
46     while(!feof(fileA))
47     {
48         char Tcode[100];
49         int sex;
50         int age;
51         int stat;
52         int degr;
53         int tyear;
54
55         //T65001001, 1,20, 1, 1
56         fscanf(fileA, "%9s,%d,%d,%d,%d", Tcode, &sex,&tyear, &degr, &stat);
57         strcpy(person[i].Tcode, Tcode);
58         person[i].sex = sex;
59         person[i].tyear = tyear;
60         person[i].stat = stat;
61         person[i].degr = degr;
62         cout << person[i].Tcode << setw(10) << person[i].sex << setw(4)
63             << person[i].tyear << setw(4) << person[i].degr << setw(4)
64             << person[i].stat << endl;
65         i++;
66     }
67     fclose(fileA);

```

```

69     avg_age = 0;    avg_expr = 0;
70     for (i = 0; i < 2; i++)
71     {
72         avg_age = avg_age + person[i].age;
73         avg_expr = avg_expr + person[i].tyear;
74     }
75
76     avg_age = avg_age / 150;
77     avg_expr = avg_expr / 150;
78
79     cout << "AVG AGE = " << avg_age << "\n";
80     cout << "AVG EXPR = " << avg_expr << "\n";
81     return 0;
82 }
83

```

```

int year;
//T65001001, 1,20, 1, 1
fscanf(fileA, "%9s,%d,%d,%d,%d", Tcode, &sex,&tyear, &degr, &stat);
strcpy(person[i].Tcode, Tcode);

```

File	Edit	Options	Encoding	Help
T65001001	,	1,20,1,1		
T65001002	,	2,22,2,2		
T65001003	,	1,21,2,2		
T65001004	,	2,22,1,2		
T65001005	,	2,23,3,2		
T65001006	,	2,19,3,2		
T65001007	,	1, 5,1,1		
T65001008	,	1,7 ,1,1		

```

char txt2[100];
fscanf(in,"%s %d %c %f %s\n", txt1, &i1, &c1, &d1, txt2);

```

```

86  /*
87  line-1·10·c·10.5·text-info-1
88  line-2·20·c·20.51·text-info-2
89  line-3·30·c·31.5·text-info-3
90  */

```

Binary File

การกับแฟ้มแบบ binary mode

- ข้อมูลที่อ่านจะไม่มี การแปลง (0xa,0xd)
- ถ้าใช้ฟังก์ชัน `fopen( const char *filename, const char *mode )` ค่าของ *mode* ต้องมี option “ b “

ตัวอย่าง

```
FILE * fp ;
```

```
fp = fopen( “c:\myfile.txt” , “rb” );
```

```
fp = fopen( “c:\myfile.txt” , “wb” );
```

```

7  int main()
8  {
9      char ch;
10     FILE *fp1 = fopen( "D:/mydoc/cos2101/DemoCOS2101/Debug/test1.txt", "rb");
11     if (fp1 == NULL) {
12         printf("cannot open file !!");
13         return 0;
14     }
15     do {
16         long pos = ftell(fp1);
17         ch = getc(fp1);
18         printf("%c \t %x \t %d \n",ch, ch , pos);
19     } while (ch != EOF);
20
21     fclose(fp1);
22
23 }
24
25

```

l	6c	0	
i	69	1	
n	6e	2	
e	65	3	rb
-	2d	4	
1	31	5	
	d	6	
	a	7	
l	6c	8	
i	69	9	
n	6e	10	
e	65	11	
-	2d	12	
2	32	13	
	d	14	
	a	15	
l	6c	16	
i	69	17	

l	6c	0	
i	69	1	
n	6e	2	
e	65	3	rt
-	2d	4	
1	31	5	
	a	6	
l	6c	8	
i	69	9	
n	6e	10	
e	65	11	
-	2d	12	
2	32	13	
	a	14	
l	6c	16	
i	69	17	
n	6e	18	
e	65	19	

File	Edit	Options	Encoding	Help																
00000000:	6C	69	6E	65	2D	31	0D	0A	6C	69	6E	65	2D	32	0D	0A		line-1..line-2..		
00000010:	6C	69	6E	65	2D	33	0D	0A	6C	69	6E	65	2D	34	0D	0A		line-3..line-4..		

	binary	
	a	23
l	6c	24
i	69	25
n	6e	26

see Instit

l	69	25
n	6e	26
e	65	27
-	2d	28
4	34	29

RUN AN...  No Config  ...

VARIABLES

Locals

```
pos = 14
ch = 10 '\n'
> fp1 = 0x75acc0
pos #2 = 14
```

> Registers

WATCH

```
ch = 10 '\n'
```

CALL STACK

[1]  PAUSED ON BREAKPOINT

main() file.c 14:1

> [2]  PAUSED

C file.c 5 X

C file.c > main()

```
2 int main()
6 if (fp1 == NULL) {
7     printf("cannot open file !!");
8     return 0;
9 }
10 do {
11     long pos = ftell(fp1);
12     ch = getc(fp1);
13     printf("%c \t %x \t %d \n",ch, ch , pos);
14 } while (ch!=EOF);
15 fclose(fp1);
16
17 }
```

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\mydoc\cos2101\apirak2\lab\lab9.file> & 'c:\Users\Apirak\.vscode\extension\ncher.exe' '--stdin=Microsoft-MIEngine-In-sufcrlfi.14m' '--stdout=Microsoft-MIE'
'--pid=Microsoft-MIEngine-Pid-svjnozm.gck' '--dbgExe=c:\msys64\ucrt64\bin\gdb'
```

```
1 6c 0
i 69 1
n 6e 2
e 65 3
- 2d 4
1 31 5
```

```
1 a 6
6c 8
```

ลำดับ operation บน binary file

1. เปิดแฟ้มข้อมูล

1. fopen ()

2. การจัดการบนแฟ้มข้อมูล

2.1. อ่าน

2.1. fread

2.2. เขียน

2.2. fwrite

2.3. การสืบค้นตำแหน่ง

2.3. fseek , ftell

3. ปิดแฟ้มข้อมูล

3. fclose()

ฟังก์ชัน fwrite

int fwrite (void **buffer*, int *size*, int *count*, FILE **stream*);

input

buffer Storage location for data
size Item size in bytes (sizeof data type)
count Maximum number of items to be written
stream Pointer to **FILE** structure

return

The **fwrite** function writes up to *count* items, of *size* length each, from *buffer* to the output *stream*.

```

118 struct.mystruct.{
119     —>int.i;
120     —>char.ch;
121     —>char.str[3];
122 };
123
124 int.main(void)
125 {
126     —>FILE.*stream;
127     —>struct.mystruct.s[10].={.0.};
128     —>if.((stream.=.fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/TEST.$$$", "wb")).==.NULL){
129     —>—>fprintf(stderr, "Cannot open output file.\n");
130     —>—>return.1;
131     —>}
132     —>for.(int.i.=.0;.i.<.10;.i++).{
133     —>—>s[i].i.=.i;
134     —>—>s[i].ch.=.'A'+.i;
135     —>—>strcpy(s[i].str, "12");
136     —>—>printf("\n%d, %c, %s", s[i].i, s[i].ch, s->str);
137     —>}
138     —>for.(int.i.=.0;.i.<.5;.i++).{
139     —>—>fwrite(&s[i], sizeof(struct.mystruct), 1, stream);
140     —>}
141
142     —>/* write struct.s to file */
143     —>fwrite(&s[5], sizeof(struct.mystruct), 5, stream);
144     —>fclose(stream); /* close file */
145     —>return.0;
146
147 }

```

```

4 struct mystruct {
5     int i;
6     char ch;
7     char str[3];
8 };
9

```

File Edit Options Encoding Help

```

00000000: 00 00 00 00 41 31 32 00|01 00 00 00 42 31 32 00 | |....A12.....B12.
00000010: 02 00 00 00 43 31 32 00|03 00 00 00 44 31 32 00 | |...C12.....D12.
00000020: 04 00 00 00 45 31 32 00|05 00 00 00 46 31 32 00 | |....E12.....F12.
00000030: 06 00 00 00 47 31 32 00|07 00 00 00 48 31 32 00 | |....G12.....H12.
00000040: 08 00 00 00 49 31 32 00|09 00 00 00 4A 31 32 00 | |....I12.....J12.

```

ฟังก์ชัน fread

```
int fread ( void *buffer, int size, int count, FILE *stream );
```

input

buffer Storage location for data

size Item size in bytes (sizeof data type)

count Maximum number of items to be read

stream Pointer to **FILE** structure

return

The **fread** function reads up to *count* items of *size* bytes from the input *stream* and stores them in *buffer*.

fread() and fwrite()

```
size_t fread(void *buffer, size_t numbytes, size_t count, FILE *a_file);
```

```
size_t fwrite(void *buffer, size_t numbytes, size_t count, FILE  
*a_file);
```

- ❑ Buffer in fread is a pointer to a region of memory that will receive the data from the file. Buffer in fwrite() is a pointer to the information that will be written to the file.
- ❑ The second argument is the size of the element; it is in bytes. For example, if you have an array of characters, you would want to read it in one byte chunks, so `numbytes` is one. You can use the `sizeof` operator to get the size of the various datatypes; for example, if you have a variable, `int x`; you can get the size of `x` with `sizeof(x)`;

Contd..

- ❑ The third argument is simply how many elements you want to read or write; for example, if you pass a 100 element array
- ❑ The final argument is simply the file pointer
- ❑ `Size_t` is an unsigned integer.
- ❑ `fread()` returns number of items read and `fwrite()` returns number of items written
- ❑ To check to ensure the end of file was reached, use the `feof` function, which accepts a `FILE` pointer and returns true if the end of the file has been reached.

```

193 struct mystruct {
194     → int i;
195     → char ch;
196     → char str[3];
197 };
198
199 int main(void)
200 {
201     → FILE *stream;
202     → //read/write where the file pointer location is at begining of file
203     → if ((stream = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/TEST.$$$", "a+b")) == NULL) {
204         → fprintf(stderr, "Cannot open output file.\n");
205         → return 1;
206     }
207
208     → while (true)
209     {
210         → struct mystruct itm;
211         → int recordno = 0;
212
213         → cout << "\nfinding record no. (0=terminate) (1-9) :: ";
214         → cin >> recordno;
215         → recordno -= 1;
216         → if (recordno < 0)
217             → break;
218
219         → fseek(stream, recordno * sizeof(struct mystruct), SEEK_SET);
220         → fread(&itm, sizeof(struct mystruct), 1, stream);
221
222         → printf("\n%d, %c, %s -> record no :: ", itm.i, itm.ch, itm.str, recordno);
223
224     }
225     → fclose(stream); /* close file */
226
227     → return 0;
228 }

```

```

1  #define _CRT_SECURE_NO_WARNINGS
2  #include <iostream>
3  #include <stdio.h>
4  #include <conio.h>
5
6  struct prod {
7      int cat_num;
8      float cost;
9  };
10
11 int main()
12 {
13     FILE *fp;
14     struct prod product;
15
16     struct prod a[3] = { {2,20.1},{4,40.1},{6,60.1} };
17     struct prod k, *p = &k;
18
19     fp = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/fread1.txt", "w+");
20     // write the entire array into the file pointed to by fp
21     fwrite(a, sizeof(product), 3, fp);
22     // prepare for reading from the beginning of the file
23     fseek(fp, SEEK_SET, 0);
24     // read from the file one product at a time
25     for (int i = 0; i < 3; i++) {
26         fread(p, sizeof(product), 1, fp);
27         printf(" product %d, cat_num=%d, cost=%f\n", i, p->cat_num, p->cost);
28     }
29     fclose(fp);
30 }

```

Rewrite the code saving data file

ตัวอย่าง

ให้นักศึกษาเขียนโปรแกรมขายสินค้าและพิมพ์ใบเสร็จรับเงิน
โดยข้อมูลในระบบมีไม่เกิน **100** รายการ

ข้อมูลขายสินค้าด้วย

- รหัสสินค้า(id) เช่น “AB001”
- ชื่อสินค้า(name) เช่น “CAFÉ”
- ราคาขาย(cost) เช่น 30.6
- จำนวนสินค้าในสต็อก (instock) เช่น 5

Menu

1. Add item to Cart
 2. Sell Product
 3. Check Stock
 4. End Program
- Please select 1 / 2 / 3/ 4 =?

กด 1

Input your product data:

id : **AB001**

name : **CAFE**

Cost : **30.6**

Stock : **5**

กด 2

Input your product id: **AB001**

Input your sell volume : **2**

name : **CAFE**

Cost : **30.6**

stock : **3**

Pay : 61.2 bath

Menu

1. Add item to Cart
2. Sell Product
3. Check Stock
4. End Program

Please select 1 / 2 / 3/ 4 =?

กด 3

product List:

Id	name	cost	stock
----	------	------	-------

AB001	CAFÉ	30.6	3
-------	------	------	---

กด 4

exit from program

```
4
5 using namespace std;
6
7 struct product {
8     char id[10];
9     char name[30];
10    float cost;
11    int instock;
12 };
13
14 int main()
15 {
16     struct product item[100];
17     int n = 0;
18     int select = -1;
19
20     while (true)
21     {
22
23         cout << " Menu" << endl;
24         cout << "1. Add item to Cart " << endl;
25         cout << "2. Sell product " << endl;
26         cout << "3. Check stock " << endl;
27         cout << "4. End program" << endl;
28         cout << "Please select 1 / 2 / 3/ 4 =? ";
29         cin >> select;
30
31         if (select == 4) {
32             break;
33         }
34         switch (select)
35         {
36             case 1:
```

single module Version

แก้ไขให้เป็น dynamic

Exercises

1. Write a C language program to read “**mark.dat**” file containing rollno, name, marks of three subjects and calculate total mark, result in grade and store same in “**result.dat**” file. (Note : Make use of fread and fwrite functions)
2. Write a C language program to read a **cust.dat** file containing meter number, name, current reading & previous reading. Read the same file. Calculate unit and total amounts according to the following rules —

Unit	rate
0-50	1.00
51-100	1.50
> 100	2.00

Store meter number ,name ,unit & amount in **master.dat** file.

END

Lister - [d:\mydoc\cos210

File Edit Options Encodi

line-1

line-2

line-3

line-4

Lister - [d:\mydoc\cos2101\apirak2\lab\lab9.file\test1.txt]

File Edit Options Encoding Help

00000000: 6C 69 6E 65 2D 31 0D 0A | 6C 69 6E 65 2D 32 0D 0A | line-1..line-2..

00000010: 6C 69 6E 65 2D 33 0D 0A | 6C 69 6E 65 2D 34 0D 0A | line-3..line-4..

```
00000000: 6C 69 6E 65 2D 31 0D 0A | line-1..line-2..
00000010: 6C 69 6E 65 2D 33 0D 0A | line-3..line-4..
```

```
1 6c 0
i 69 1
n 6e 2
e 65 3
_ 2d 4
1 31 5

1 a 6
i 6c 8
n 69 9
e 6e 10
_ 65 11
2 2d 12
3 32 13

1 a 14
i 6c 16
n 69 17
e 6e 18
_ 65 19
3 2d 20
4 33 21

1 a 22
i 6c 24
n 69 25
e 6e 26
_ 65 27
4 2d 28
34 29
```

```
7 int main()
8 {
9     char ch;
10    FILE *fp1 = fopen( "D:/mydoc/cos2101/DemoCOS2101/Debug/test1.txt", "rt");
11    if (fp1 == NULL) {
12        printf("cannot open file !!");
13        return 0;
14    }
15    do {
16        long pos = ftell(fp1);
17        ch = getc(fp1);
18        printf("%c \t %x \t %d \n",ch, ch , pos);
19    } while (ch != EOF);
20
21    fclose(fp1);
22
23 }
```

```
line-1
line-2
line-3
line-4
```

<https://www.jdoodle.com/online-compiler-c++14>

ฟังก์ชัน ftell

long ftell(FILE **stream*);

input

stream Pointer to **FILE** structure

return

returns the current file position. The value returned by **ftell** may not reflect the physical byte offset for streams opened in text mode

ฟังก์ชัน fseek

```
int fseek( FILE *stream, long offset, int origin );
```

input

stream Pointer to **FILE** structure

offset Number of bytes from *origin*

origin Initial position

- **SEEK_CUR** Current position of file pointer
- **SEEK_END** End of file
- **SEEK_SET** Beginning of file

return

0 if successful,

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FILE *fp ;

fp = fseek(fp , 0 , SEEK_SET); */* go to top of file */*

fp = fseek(fp , 5 , SEEK_SET); */* go to 5 byte from top file */*

fp = fseek(fp , 0 , SEEK_END); */* go to button of file */*

fp = fseek(fp , - 5 , SEEK_END); */* go back 5 byte from button */*

fp = fseek(fp , 5 , SEEK_CUR); */* skip next 5 byte from cur pos */*

fp = fseek(fp , -5 , SEEK_CUR); */* go back 5 byte from curs pos*/*

fseek

```
5  int main()
6  {
7      char ch;
8      FILE *fp1 = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/test1.txt", "rt");
9      if (fp1 == NULL) {
10         printf("cannot open file !!");
11         return 0;
12     }
13
14     do {
15         long pos1;
16         printf("seek position (-1 exit) : ");
17         scanf("%d", &pos1);
18         if (pos1 < 0) {
19             break;
20         }
21         fseek(fp1, pos1, SEEK_SET);
22         long pos2 = ftell(fp1);
23         ch = getc(fp1);
24         printf("%c \t %x \t %d \n", ch, ch, pos2);
25     } while (true);
26
27     fclose(fp1);
28
29 }
30
```

NOTE: เพิ่มแบบ text mode

ลักษณะของเพิ่มแบบ text โหมด

1. มี CTRL-Z บอกเครื่องหมายจบเพิ่ม
2. เครื่อง carriage return–linefeed (0xa,0xd) ถูกแทนด้วย linefeed (0xd)
3. การใช้คำสั่งค้นหาตำแหน่งข้อมูล (fseek,ftell) อาจคลาดเคลื่อน
4. ถ้าใช้ฟังก์ชัน `fopen( const char *filename, const char *mode )` ค่าของ *mode* ต้องมี option “ t “ หรือไม่ option “ b ”

ตัวอย่าง

```
FILE *fp ;
```

```
fp = fopen( “c:\myfile.txt” , “rt” );    → Text File
```

```
fp = fopen( “c:\myfile.txt” , “rb” );    → Binary File
```

Basic File Steps to Handle File

- Opening a file
- Reading data from a file
- Writing data to a file
- Closing a file

```
7  int main()
8  {
9      char ch;
10     FILE *fp1 = fopen( "D:/mydoc/cos2101/DemoCOS2101/Debug/test1.txt", "rt");
11     if (fp1 == NULL) {
12         printf("cannot open file !!");
13         return 0;
14     }
15     do {
16         long pos = ftell(fp1);
17         ch = getc(fp1);
18         printf("%c \t  %x \t %d \n",ch, ch , pos);
19     } while (ch != EOF);
20
21     fclose(fp1);
22
23 }
```


Example: Merge two files

```
#include <stdio.h>
int main()
{
    FILE *fileA, /* first input file */
        *fileB, /* second input file */
        *fileC; /* output file to be created */
    int num1, /* number to be read from first file */
        num2; /* number to be read from second file */
    int f1, f2;

    /* Open files for processing */
    fileA = fopen("class1.txt", "r");
    fileB = fopen("class2.txt", "r");
    fileC = fopen("class.txt", "w");
```

```
/* As long as there are numbers in both files, read and compare
numbers one by one. Write the smaller number to the output file
and read the next number in the file from which the smaller
number is read. */
```

```
f1 = fscanf(fileA, "%d", &num1);
f2 = fscanf(fileB, "%d", &num2);
```

```
while ((f1!=EOF) && (f2!=EOF)) {
    if (num1 < num2) {
        fprintf(fileC, "%d\n", num1);
        f1 = fscanf(fileA, "%d", &num1);
    }
    else if (num2 < num1) {
        fprintf(fileC, "%d\n", num2);
        f2 = fscanf(fileB, "%d", &num2);
    }
    else { /* numbs are equal: read from both files */
        fprintf(fileC, "%d\n", num1);
        f1 = fscanf(fileA, "%d", &num1);
        f2 = fscanf(fileB, "%d", &num2);
    }
}
```

```
while (f1!=EOF){/* if reached end of second file, read
    the remaining numbers from first file and write to
    output file */
    fprintf(fileC,"%d\n", num1);
    f1 = fscanf(fileA, "%d", &num1);
}
while (f2!=EOF){ if reached the end of first file, read
    the remaining numbers from second file and write
    to output file */
    fprintf(fileC,"%d\n", num2);
    f2 = fscanf(fileB, "%d", &num2);
}

/* close files */
fclose(fileA);
fclose(fileB);
fclose(fileC);
return 0;
} /* end of main */
```